Shivam Maurya

AI Engineer | LLM Applications, AI Agents, RAG Systems and Intelligent Automation

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SUMMARY

AI Engineer who builds real-world LLM-powered products, AI agents, and agentic workflows beyond basic API integrations. Built a 5-phase LangGraph credit-appraisal agent with tool calling, RAG (Pinecone, LlamaParse), human-in-the-loop review, and multi-model orchestration (Gemini, Llama 3.3 on Groq, Ollama), plus a 9-model diagnostic platform orchestrated by an LLM with hallucination safeguards. Takes a requirement from architecture, model selection, data, retrieval, agents, and APIs through evaluation and deployment, with a focus on reliability, accuracy, latency, and cost. Amazon ML Challenge 2026 Top 500 (All India), Hackathon 2026 Runner-Up, 15+ merged open-source pull requests.

TECHNICAL SKILLS

LLM and Agentic AI: LLM Applications, AI Agents, Agentic Workflows, Multi-Agent Orchestration, Tool Calling, ReAct Pattern, LangGraph, LangChain, Prompt Engineering, Context Management, Human-in-the-Loop, Hallucination Mitigation, LLM Output Evaluation

RAG and Retrieval: Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone), Embeddings, Document Parsing (LlamaParse, PyMuPDF, Tesseract OCR), Web Search Integration (Tavily)

LLM APIs and Models: Google Gemini API, Groq API (Llama 3.3), Ollama (Gemma), Sarvam AI Speech-to-Text and Text-to-Speech, Model Selection, Model Benchmarking

Languages and Backend: Python, SQL, C++, Bash, FastAPI, Flask, REST APIs, WebSockets, Pydantic, Streamlit, React, SQLite

Data and Machine Learning: Data Pipelines, Feature Engineering, Pandas, NumPy, scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Keras, ONNX Runtime, SHAP, LIME, Cross-Validation

Deployment and Tools: Docker, Git, GitHub, Linux, CI/CD, pytest, Microservices, API Integration, Automation Systems

EXPERIENCE

Physical AI Intern | Nextup Robotics | Remote | Aug 2026 - Oct 2026

- Built a synchronized multimodal data-generation pipeline (20 calibrated RTX cameras plus tactile contact sensors across 7 rigid bodies per robot) producing vision and tactile datasets for NVIDIA Cosmos foundation models.
- Scaled a simulation to 30 UR10e robots and 20 cameras at 40-50 FPS on a single consumer GPU workstation (RTX 5060 Ti 16GB, about 8.2 GiB RAM) through profiling and buffer tuning.
- Orchestrated a 3-arm, multi-step workflow across four stations using custom ROS 2 action servers and state machines, and tuned FastDDS middleware for zero-packet-drop synchronization between MoveIt 2 and Isaac Sim.
- Authored 7 engineering guides covering workcell setup, calibration, and headless CI deployment.

Software Lead | Centre of Excellence for AI & Robotics, Army Institute of Technology, Pune | Jun 2024 - Present

- Led software for the ABU Robocon national competition robot (All India Rank under 20), deploying real-time CNN perception into navigation decision loops.
- Built an ONNX export pipeline and a JSON-configurable, hot-swappable inference backend that enables runtime model swaps and A/B testing with zero downstream code changes, reaching 30+ FPS on CPU-only hardware.
- Trained and validated 4+ ML architectures (CNN, Xception transfer learning, Random Forest, XGBoost) using stratified cross-validation, data augmentation, and automated hyperparameter sweeps.
- Directed a 100+ member research cell; organized hackathons, workshops, and project sprints across 5+ teams.

Open-Source Contributor | ROS 2 Ecosystem (navigation2, ros2_control, ros2_controllers) | Jun 2025 - Present

- Authored 15+ merged pull requests in widely used robotics repositories, working with global maintainers through code review.
- Diagnosed a regression in controller manager initialization by analyzing backtraces, fixed cross-repository CI build failures, and added static type checking (ament_mypy) to improve code reliability.

PROJECTS

CREDI-MITRA: Autonomous AI Credit Appraisal Agent | LangGraph, RAG, Pinecone, Gemini, Groq, XGBoost, Streamlit | Hackathon 2026 Runner-Up | github.com/ShivamMaurya14/CREDI-MITRA

- Designed a 5-phase stateful LangGraph ReAct agent with tool calling across document analysis, live web research, feature extraction, ML scoring, and automated PDF Credit Appraisal Memo generation, with human-in-the-loop review at every critical decision stage.
- Selected models by task with a dual-LLM architecture: Llama 3.3 and Gemini for orchestration, Gemini 2.5 Pro for deep analysis.

- Built a RAG pipeline with LlamaParse and Pinecone that reduced hallucinations by 85%, and integrated Tavily search for litigation (NCLT filings, RBI penalties) and news-sentiment cross-verification.
- Deployed an XGBoost scoring tool (97% accuracy) predicting loan approval, credit limit, and interest rate.

MedSynapse: LLM Orchestration of 9 Diagnostic Models | FastAPI, React, Groq, Ollama, PyTorch, TensorFlow | Avishkar 2026 State Level Round 2, IIIT Pune Thinking Machines 2025 Finalist | github.com/ShivamMaurya14/MedSynapse

- Architected an LLM orchestration layer over 9 specialized ML and deep learning models (up to 99.25% accuracy) that synthesizes cross-module correlations and generates calibrated clinical reports with hallucination safeguards.
- Built a multimodal ingestion pipeline for clinical text, audio, X-rays, MRI, fundus images, and lab PDFs using Tesseract OCR and PyMuPDF, backed by a SQLite feature store.
- Implemented fused explainability (SHAP, LIME, Grad-CAM, DiCE) with SHAP-LIME agreement scoring for confidence-aware outputs, and a clinician review and approval workflow.
- Delivered as a full-stack FastAPI and React application using Groq and local Ollama (Gemma) models.

MedSarthi: Multilingual Voice Health Assistant | Gemini 2.0 Pro, Sarvam AI, FastAPI, TensorFlow | github.com/ShivamMaurya14/MedSarthi

- Built a voice pipeline (browser recording, speech-to-text, Gemini LLM conversation, text-to-speech) supporting Hindi, English, and other Indic languages, deployable as a single FastAPI process with no external infrastructure.
- Integrated 4 ML and deep learning diagnostic models locally and added AI triage with urgency classification, an AI-generated triage report dashboard for doctors, and appointment scheduling.

Edge Voice Activator for Home Automation | ESP32-S3, TFLite, Vosk ASR, Zenoh, WebSockets | Smart India Hackathon 2026 (ISRO PS-26172)

- Designed a 3-stage cascading intent parser (Regex, then Fuzzy Levenshtein matching, then ML fallback) that tries cheaper methods first, reaching over 97% intent accuracy on noisy speech-recognition output.
- Optimized latency across the stack: INT8-quantized 111KB wake-word model on-device, and Zenoh transport at about 1-2 ms on LAN versus about 14 ms for WebSockets.

RealFake Classifier: Real-Time Perception Pipeline | ONNX Runtime, YOLOv8, TensorFlow, ROS 2, Flask | ABU Robocon | github.com/ShivamMaurya14/RealFake-Classifier-Robocon

- Benchmarked three inference backends on identical input: ONNX Runtime 3.89 ms (257 FPS), YOLOv8 about 18.5 ms (about 54 FPS), TensorFlow 22.18 ms (45.1 FPS), and shipped the fastest as the default with hot-swappable alternatives.
- Built a decision engine that maps model output to robot actions and a Flask dashboard with live telemetry and backend switching.

Credit Risk Intelligence System | Random Forest, XGBoost, SHAP, FastAPI | github.com/ShivamMaurya14/CreditRisk-Predictor

- Built an end-to-end pipeline on 307,511 loan applications with 125 engineered features, achieving ROC-AUC 0.8866 and 91.46% default recall, and optimized the decision threshold for net financial benefit.
- Kept research and production in sync through a shared calibration file and served predictions with SHAP reason codes via FastAPI.

Industrial IoT Predictive Maintenance Platform | FastAPI, Streamlit, Random Forest, ESP32 | github.com/ShivamMaurya14/Predictive-Maintenance-System | Live demo: pms-shivam-maurya.streamlit.app

- Built a decoupled microservice system with a FastAPI ingestion API for live ESP32 sensor data and a Streamlit dashboard, with over 98% failure-prediction accuracy and four operating modes.

EDUCATION

Bachelor of Engineering (B.E.), Automation and Robotics Engineering | Army Institute of Technology, Pune, India | Jun 2024 - May 2028 (Expected) | CGPA: 8.06 / 10

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Probability and Statistics, Data Structures and Algorithms, Database Management Systems (SQL), Object-Oriented Programming

HONORS & ACHIEVEMENTS

- Top 500 All India, Amazon ML Challenge 2026 (F0.5 score 0.9602)
- Runner-Up, Hackathon 2026 (CREDI-MITRA)
- State Level Round 2, Avishkar Maharashtra Inter-University Research Convention 2026 (MedSynapse)
- Top Submission, CODECURE Hackathon 2025, IIT BHU (antibiotic resistance prediction, 99.27% accuracy)
- Finalist, IIIT Pune Thinking Machines 2025 (MedSynapse)
- All India Rank under 20, ABU Robocon 2025/26 (Lead Software Architect)

Languages: English (Professional Working Proficiency), Hindi (Native)